Purely Analytic Closure of the Final Spherical-Shell Residual

Abstract

This note assembles the all-frequency retained-remainder estimate after the exact owner subtraction. It proves that the final spherical-shell residual is empty without invoking an interval certificate, an executable truth table, an aggregate tail argument, or a frequency split. The structural, angular-block, and scalar-positivity modules cover one closed superset of the entire residual.

1 Notation and analytic inputs

Let

$$A_{\rho,1} := \{x \in \mathbb{R}^3 : \rho < |x| < 1\}, \qquad W(\rho, K) := \frac{2(1-\rho^3)K^3}{9\pi},$$

and let $N_D(A_{\rho,1}, K^2)$ count Dirichlet eigenvalues strictly below K^2. Put

$$\rho_c := \frac{1}{1+2\pi}, \qquad k_{11}(\rho) := \sqrt{\frac{\pi^2}{(1-\rho)^2} + 132}.$$

The preceding analytic owner lemmas and the exact set subtraction give

$$\mathcal{D}_{20} = \left\{(\rho, K) : \rho_c \le \rho < \frac{39}{50}, \quad k_{11}(\rho) < K < U(\rho)\right\}. \tag{1}$$

On this interval $U(\rho) = K_0(\rho)$, although the formula for the upper wall will not be needed below.

We use the shell action

$$G_a(z) := \frac{1}{\pi}\left(\sqrt{a^2 - z^2} - z \arccos\frac{z}{a}\right) \quad (0 \le z < a), \qquad G_a(z) := 0 \quad (z \ge a),$$

$$A_{\rho,K}(z) := G_K(z) - G_{\rho K}(z),$$

and the strict lattice proxy

$$P(\rho, K) := \sum_{\substack{\nu = \ell + 1/2 < K \\ \ell \in \mathbb{N}_0}} 2\nu \,\#\left\{n \in \mathbb{N} : n < A_{\rho,K}(\nu) + \frac{1}{4}\right\}. \tag{2}$$

The analytic Bessel-phase reduction proved earlier in the main argument is

$$N_D(A_{\rho,1}, K^2) \le P(\rho, K). \tag{3}$$

Put

$$M_c(\rho, K) := \frac{(1-\rho^2)K^2}{6} - \frac{1}{2}.$$

Let

$$N := \#\left\{n \geq 1 : n - \frac{1}{4} < \frac{(1-\rho)K}{\pi}\right\}. \tag{4}$$

Parts VII–VIII of the detailed analytic supplement combine the retained radial remainder with one disjoint left block per active layer and give the single direct estimate

$$W(\rho, K) - P(\rho, K) > \mathcal{H}(\rho, K) - \frac{(1-\rho^2)K^2}{6} + \frac{N}{2}. \tag{5}$$

On $\rho_c \leq \rho \leq 39/50$ and $K \geq k_{11}(\rho)$, one has $(1-\rho)K/\pi > 1$, hence $N \geq 1$. Part IX gives on the same unbounded domain

$$\mathcal{H}(\rho, K) - M_c(\rho, K) > 0. \tag{6}$$

The all-frequency scope in (6) is part of that module: its scalar gap has positive K-derivative whenever $K > 12$, one has $k_{11}(\rho) > 241/20 > 12$, and the gap is strictly positive on the base curve $K = k_{11}(\rho)$. Thus the proof integrates upward from the base curve to every finite $K \geq k_{11}(\rho)$; it uses no upper-frequency bound.

2 All-frequency retained-remainder closure

Theorem 1 (All-frequency middle-ratio theorem)**.** *For*

$$\rho_c \leq \rho \leq \frac{39}{50}, \qquad K \geq k_{11}(\rho),$$

one has

$$N_D(A_{\rho,1}, K^2) < W(\rho, K).$$

Proof. Since $N \geq 1$, equations (5) and (6) give

$$W - P > \mathcal{H} - M_c > 0.$$

The phase reduction (3) completes the proof. Strictness includes both closed ratio faces, the base face $K = k_{11}(\rho)$, and every common radial–angular proxy wall. □

3 Direct ownership of the exact residual

Proposition 2 (Analytic closure of $\mathcal{D}_{20}$)**.** *Define the closed all-frequency theorem domain*

$$\mathcal{R}_{\mathrm{rr}} := \left\{(\rho, K) : \rho_c \leq \rho \leq \frac{39}{50}, \quad K \geq k_{11}(\rho)\right\}.$$

Then $\mathcal{D}_{20} \subset \mathcal{R}_{\mathrm{rr}}$, the strict Pólya inequality holds throughout $\mathcal{D}_{20}$, and the successor residual is empty.

Proof. If $(\rho, K) \in \mathcal{D}_{20}$, then (1) gives

$$\rho_c \leq \rho < \frac{39}{50}, \qquad k_{11}(\rho) < K < U(\rho).$$

After weakening the strict upper ratio and lower frequency inequalities, these conditions place (ρ, K) in $\mathcal{R}_{\mathrm{rr}}$. Therefore Theorem 1 proves every point of $\mathcal{D}_{20}$, and

$$\mathcal{D}_{20} \setminus \mathcal{R}_{\mathrm{rr}} = \varnothing.$$

The already owned faces $\rho = 39/50$, $K = k_{11}(\rho)$, and $K = U(\rho)$ remain outside $\mathcal{D}_{20}$, even though the stronger theorem also covers the first two. The face $\rho = \rho_c$ remains in $\mathcal{D}_{20}$ whenever both strict frequency inequalities hold. No artificial frequency face is present. □

Logical dependency. After Parts I–VI identify $\mathcal{D}_{20}$, the closure uses only the structural, angular-block, and scalar-positivity modules in Parts VII–IX. It does not use an aggregate-tail module, a frequency split, the former 10,580-box compact certificate, the 1,286-box aggregate certificate, or the 63-state executable subtraction table.